

# Corrective Action Report — CAR-2025-007

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| Field                          | Value                                                                                                                                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document ID                    | CAR-2025-007                                                                                                                                                                       |
| Issue Date                     | 2025-10-02                                                                                                                                                                         |
| Author                         | F. Ahuja, Engineering Documentation Auditor; reviewed by D. Pham, Compliance Lead                                                                                                  |
| CAR Status                     | <b>Open</b>                                                                                                                                                                        |
| Subject                        | GE Renewable Energy OEM Manual References Pre-2023 Firmware Inspection Cadence; Field Procedures for WIN-10016 and WIN-10058 Currently Have No Authoritative Maintenance Reference |
| Affected Region / Site / Asset | NW-PowerPool / NW-PowerPool-WIND-014 / WIN-10016, WIN-10058                                                                                                                        |
| Severity                       | High                                                                                                                                                                               |
| Owner                          | E. Linhart, Principal Engineer — Rotating Equipment; co-owner G. Calderon, OEM Liaison                                                                                             |
| Original Target Close Date     | 2025-12-31 (missed; documentation not yet received)                                                                                                                                |
| Revised Target Close Date      | 2026-06-30 (dependent on OEM delivery)                                                                                                                                             |
| Distribution                   | Engineering Standards Committee; NW-PowerPool Plant Manager; Documents Control; Procurement; Warranty Administration                                                               |

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## 1. Finding

GE Renewable Energy OEM Service Manual (filed **VTM-GE-2023**, internal revision tag "2.4") is the currently controlled document for service on the 2.x-series wind turbines deployed at NW-PowerPool-WIND-014, including WIN-10016 and WIN-10058. In June 2023, GE deployed a coordinated firmware refresh (Bulletin GE-SB-2023-014) to the affected fleet that materially changed:

1. Pitch-system control-loop timing (5 ms control loop → 2 ms control loop).

2. Yaw-system fault-tolerance threshold logic (legacy single-fault detection → dual-fault voting).
3. Drivetrain over-temperature trip setpoint (90 °C generator stator → 87 °C, with new pre-trip warning at 82 °C).
4. SoC (state-of-charge) battery management for emergency pitch-shutdown.

The VTM-GE-2023 manual was published in February 2023 — four months before the firmware refresh. Inspection cadences and procedural steps in §4 ("Periodic Inspection — Operating Asset") are calibrated to pre-refresh setpoints and signal timings. GE committed in writing (correspondence dated 2024-01-19) to deliver a Rev. 3 manual reflecting the firmware refresh. As of this CAR issue date, **no Rev. 3 has been received**. The follow-up commitment was for 2024-Q3 delivery; subsequent commitment dates of 2025-Q1, 2025-Q2, and 2025-Q4 have all lapsed.

Field maintenance is therefore being performed against a manual whose inspection procedures cannot reconcile with deployed asset behavior. Two specific operational signatures have emerged.

## 2. Evidence

### 2.1 Procedural-deployed-behavior mismatches

| Manual Procedure                             | VTM-GE-2023 Specification                   | Post-Refresh Reality                                                                                                            | Impact                                                  |
|----------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| §4.2.3 Pitch loop test                       | Inject step input, observe ≤120 ms response | Deployed loop responds in ≤45 ms; manual procedure flags as out-of-spec (false positive)                                        | Field technicians log unnecessary "anomaly" tickets     |
| §4.5.1 Yaw fault simulation                  | Single fault injection; expect alarm        | Dual-fault voting requires simultaneous injection; manual procedure cannot trigger alarm; technicians report "no alarm" as pass | False-negative; actual yaw fault tolerance not verified |
| §4.7.2 Generator over-temperature simulation | Inject 90 °C signal; expect trip            | Trip occurs at 87 °C — earlier than expected, manual procedure produces a "premature trip" finding                              | False anomaly flag                                      |
| §4.8.4 SoC test                              | Procedure assumes 8.5 V emergency floor     | Refresh moved emergency floor to 9.2 V                                                                                          | Test fails on healthy assets                            |

## 2.2 Linked outages

- OUT-2025-0345 (2025-04-27, WIN-10016, soc\_drift, 219 h, 2,506.8 MWh lost) — root cause hypothesized as SoC trip threshold drift; investigation hampered because the §4.8.4 procedure produces false fails on the deployed firmware. Investigation took 38 days to close, of which 12 days were the documentation-misalignment delay.
- OUT-2025-0433 (2025-05-02, WIN-10058, soc\_drift, 227 h, 2,506.8 MWh lost) — same fleet, same uncertainty about SoC-test interpretation.
- OUT-2025-0336 (2025-04-01, WIN-10016, overheating, 175 h, 1,932.5 MWh lost) — generator stator over-temperature; manual specifies different setpoint than deployed.
- WIN-10016 has accumulated 16 forced outages in FY2025; WIN-10058 has accumulated 12. Both are on the same site (NW-PowerPool-WIND-014). Both rely on VTM-GE-2023.

## 2.3 OEM correspondence chain

| Date       | Direction     | Substance                                                             |
|------------|---------------|-----------------------------------------------------------------------|
| 2023-06-14 | GE → Cascadia | Bulletin GE-SB-2023-014 announcing firmware refresh                   |
| 2024-01-19 | GE → Cascadia | Acknowledgment that manual Rev. 3 is required; commitment for 2024-Q3 |
| 2024-09-11 | Cascadia → GE | Request status of Rev. 3                                              |
| 2024-10-02 | GE → Cascadia | Revised commitment 2025-Q1                                            |
| 2025-03-31 | Cascadia → GE | Status request; received no reply                                     |
| 2025-06-13 | Cascadia → GE | Escalation to OEM account manager                                     |
| 2025-06-27 | GE → Cascadia | Revised commitment 2025-Q4                                            |
| 2025-10-01 | Cascadia → GE | Status request — no reply as of CAR issue                             |

## 2.4 Workaround currently in place

Cascadia Engineering issued **Field Bulletin FB-2024-007** ("Interim Inspection Procedure for Refresh-2023 GE Fleet") on 2024-03-15. FB-2024-007 supersedes specific sections of VTM-GE-2023 but has not been integrated into the controlled-documentation system. Field technicians are nominally to apply FB-2024-007 on top of VTM-GE-2023; in practice, audit shows only 64% of inspections cite FB-2024-007 in their work-order documentation.

## 3. Root Cause Hypothesis

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- **Primary:** GE has not delivered a firmware-current manual revision. The accountability for delivery rests with GE; the accountability for managing the gap rests with Cascadia OEM Liaison.
- **Contributing:** FB-2024-007 is an unintegrated workaround. It is not a controlled-document object in the engineering documentation system; it exists as a PDF in an Engineering SharePoint. It does not appear in CMMS work-order pre-population.
- **Contributing:** No escalation path was activated when GE missed the 2024-Q3 delivery commitment. The Vendor Performance Review process (cross-reference Procurement Policy PP-2023-04 §9) would have triggered a notice-of-non-conformance at 90 days past commitment.

## 4. Conflict With Existing Guidance

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- **VTM-GE-2023** is the controlled document; it conflicts in four specific procedures with the firmware deployed since June 2023.
- **FB-2024-007** is an uncontrolled workaround; its inconsistent application is itself a documentation-control deficiency.
- **Cascadia Engineering Documentation Standard EDS-2022-01 §4.2** requires that all controlled documents reflect deployed asset behavior; this requirement is currently unsatisfiable absent OEM delivery.

Unlike CAR-2025-002 (Siemens Gamesa) and CAR-2025-005 (ABB), this conflict is not between OEM manual and internal standard. It is between OEM manual and physical reality of the OEM's own deployed firmware.

## 5. Recommended Corrective Action

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1. Issue formal notice-of-non-conformance to GE under PP-2023-04 §9 within 30 days, citing 18+ months of missed manual-delivery commitments.

2. Reissue FB-2024-007 as a controlled document **CD-FB-2024-007** in the engineering documentation system, with mandatory CMMS pre-population on any GE 2.x-series work order.
3. Suspend §4.2.3, §4.5.1, §4.7.2, and §4.8.4 procedures from VTM-GE-2023 with formal procedure-suspension notice. Replace with the FB-2024-007 procedures pending Rev. 3 delivery.
4. Add a Vendor-Performance contractual remedy for delayed documentation in the next renewal of the GE service relationship.
5. Audit all closed GE work orders 2024-01-01 to date for FB-2024-007 application; re-perform where evidence is missing.

## 6. Action Owner & Timeline

| Sub-action                                | Owner                                 | Target Date                |
|-------------------------------------------|---------------------------------------|----------------------------|
| Non-conformance notice to GE              | C. Rourke (Procurement) + G. Calderon | 2025-11-30 — <b>missed</b> |
| FB-2024-007 to controlled-document status | F. Ahuja (Doc Control)                | 2025-12-31 — <b>missed</b> |
| CMMS pre-population update                | R. Vasquez                            | 2026-02-28                 |
| Audit of closed GE work orders            | Reliability & Compliance              | 2026-03-31                 |
| Contractual remedy in next GE renewal     | Legal + Procurement                   | At next renewal (2026-Q4)  |

## 7. Implementation Status

**Open.** The non-conformance notice has not been issued (held pending review of the broader GE service relationship by Procurement). FB-2024-007 remains uncontrolled. WIN-10016 has experienced two additional forced outages since this CAR was opened (OUT-2025-1495, OUT-2025-1601 — non-NW-PowerPool-WIND-014 cross-reference for site-internal cascade); the SoC-related uncertainty persists.

**Cross-references:** CAR-2025-001 (deferred PMs for WIN-10016 / WIN-10058 — compounding effect); CAR-2025-009 (escalation policy: WIN-10016 at 16 outages should have been engineering-reviewed at 3); FB-2024-007 (uncontrolled workaround); VTM-GE-2023; Bulletin GE-SB-2023-014; PP-2023-04 §9 (vendor performance); EDS-2022-01 §4.2 (documentation standard); OIR-2025-003 (Period A); RCA-2025-001 (WIN-10058 root cause).

*End of report.*